STGNI: Spatio-Temporal Graph Node Importance for Software-Defined Networking Traffic Prediction

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***Abstract*—Network traffic prediction enables SDN controllers to optimize traffic engineering and load balancing, thereby mitigating congestion and faults while improving resource utilization. While most studies focus on spatio-temporal correlations in traffic data, in practical networks link capacities differ, resulting in unequal influences among nodes. We define this unequal contribution as node importance. To model this phenomenon, we propose STGNI (Spatio-Temporal Graph Node Importance for Software-Defined Networking Traffic Prediction), a model comprising three core modules: DNM (Data Normalization Module), NIM (Node Importance Module), and STDM (Spatio-Temporal Dependency Module). Experiments on the Abilene and GEANT datasets demonstrate the effectiveness of STGNI, outperforming eight baseline methods across three evaluation metrics and multiple prediction horizons (1–5 steps).**

***Index Terms*—Network Traffic Prediction, Graph Node Importance, Spatio-Temporal Correlation, Software-Defined Networking**

I. Introduction

With the rapid growth of Internet technologies and online services, network traffic is increasing exponentially, posing serious challenges to traditional architectures. Emerging applications such as industrial robotics and autonomous systems further demand real-time and reliable transmission. SDN (Software-Defined Networking) addresses these challenges by decoupling the control and data planes, enhancing flexibility, programmability, and adaptability to dynamic service demands [1]. In SDN, the controller monitors network status and makes real-time and predictive decisions on traffic management, including topology adjustment, routing optimization, and load balancing. Accurate traffic prediction is therefore essential for efficient operation and intelligent control, enabling resource optimization, congestion mitigation, and reliable service delivery [2, 3].

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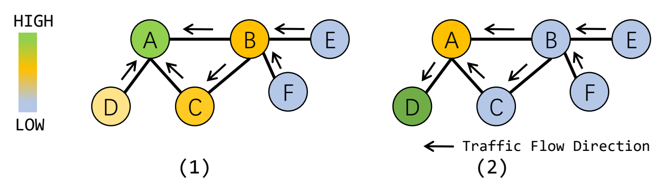

Fig. 1. A Toy Traffic Network Example with 6 Nodes. Node Colors Indicate Contribution Levels to Specific Nodes.

Existing methods fall into two main categories: traditional statistical methods [4, 5] and deep learning methods [6, 7]. To better capture the spatio-temporal relationships in network traffic, many researchers have incorporated graph convolutional networks [8] and attention mechanisms [9], leading to more accurate traffic predictions [7, 10]. However, despite these advancements, a key challenge remains: in real-world networks, the traffic-bearing capacity of links varies, and nodes contribute differently to each other. For instance, a node's importance is determined by how much traffic it passes to other nodes. similar to how node interactions and topology critically affect consensus performance in multi-agent systems [11]. Moreover, network issues such as congestion, interruptions, or equipment failures can cause outliers in traffic data, complicating predictions.

To facilitate analysis, we assume uniform traffic distribution among network nodes. As shown in Figure 1 (1), when focusing on node A, nodes B and C exhibit greater importance than node D, since D only carries its own traffic while B and C also aggregate information from nodes E and F. Similarly, from the perspective of node D (Figure 1 (2)), node A holds significantly higher importance than other nodes. We refer to this implicit relational property as node importance, which plays a crucial role in traffic prediction.

In this work, we propose a traffic prediction model based on node importance, addressing two key challenges:

1) How to normalize data across nodes with varying traffic volumes.

2) How to effectively quantify node importance.

To tackle these challenges, we propose STGNI (Spatio-Temporal Graph Node Importance for Network Traffic Prediction). To address the first challenge, we introduce a DNM (Data Normalization Module) that normalizes raw traffic data using the median and interquartile range of each node, reducing the impact of outliers. For the second challenge, we design a NIM (Node Importance Module), which computes the importance of surrounding nodes via graph attention mechanisms [12] and adjusts it based on the network topology. Furthermore, to capture spatial and temporal dependencies, we leverage normalized data and node importance within a dual-layer LSTM with linear layers, referred to as the STDM (Spatio-Temporal Dependencies Module). Experimental results demonstrate that our method effectively determines the importance of each node and enhances the model's internal spatio-temporal relationships, leading to more accurate traffic predictions.

The main contributions of this work are summarized as follows:

1) We propose STGNI, a novel method for network traffic prediction that explicitly models the spatio-temporal dependencies in node importance through three specialized modules: DNM, NIM, and STDM.

2) The DNM addresses data heterogeneity by performing node-wise normalization using median and interquartile range, effectively handling traffic variations and outliers. The NIM quantifies node importance through graph attention mechanisms and linear transformations, while the STDM captures complex spatio-temporal patterns using a dual-layer LSTM architecture.

3) Extensive experiments on the Abilene and GEANT datasets demonstrate the effectiveness of STGNI and validate the contribution of each component.

The remainder of this paper is organized as follows: Section II reviews related work. Section III formulates the network traffic prediction problem. Section IV details the STGNI architecture. Section V presents experimental results and analysis. Finally, Section VI concludes the paper.

II. Related work

In network traffic prediction, the primary objective is to forecast future traffic volumes, typically measured in bits or bytes, which represent data transmitted through computer networks. This data encompasses various traffic types, including web requests, email transmissions, file transfers, multimedia streams, and database queries [3].

Current research employs two main methodological approaches: traditional statistical methods and deep learning-based techniques. Statistical methods, such as ARIMA [4, 5], model traffic as time series data under assumptions of stationarity and independence, but exhibit limitations in capturing complex nonlinear relationships. In contrast, deep learning methods leverage machine learning [13, 14] or deep neural networks [6, 15] to automatically extract spatial and temporal features from traffic data, demonstrating superior performance in modeling nonlinear patterns. Recent advances focus on integrating spatio-temporal modeling techniques to better capture network traffic characteristics. Such spatio-temporal approaches have shown effectiveness not only in network traffic prediction but also in related domains like traffic flow forecasting [7, 10, 16, 17].

III. Preliminary

Network traffic prediction is represented using an undirected graph $G = (V, E, A)$, where $V = \{v_i | i \in N\}$ denotes the nodes in the network, $N = |V|$ represents the number of nodes, E represents the edges connecting two nodes, and $A = \{a_{ij} | i, j \in N\}$ denotes the adjacency matrix, which represents the connection between node i and node j. Here, $a_{ij} \in \{0, 1\}$, a value of 1 indicates a connection between the two nodes, while 0 indicates no connection.

In the network traffic prediction problem, the network traffic is expressed as $\mathrm{X} = \{x_{i,t} \mid i \in N, t \in R\}$,

where $x_{i,t}$ represents the traffic data of node i at time step t. We use historical traffic data X and the adjacency matrix A to predict future network traffic over T time steps (where $T \leq 5$), denoted as $X_T^{'}$. The goal is to find a function f that maps X and A to $X_T^{'}$:

$$X_T^{'} = f(X, A). \tag{1}$$

Here, $X_T^{'} = \{x_{i,t+T}^{'} \mid i \in N, t \in R, T \in \{1, 2, 3, 4, 5\}\}$.

IV. Proposed Model

The STGNI model, illustrated in Figure 2, comprises three key modules: the Data Normalization Module (DNM), the Node Importance Module (NIM), and the Spatio-Temporal Dependency Module (STDM).

A. Data Normalization Module.

The DNM normalizes historical traffic data using RobustScaler [18], which processes each node individually based on its median and interquartile range. While analyzing and processing the dataset, we encountered the following challenges:

1) Significant Variations in Traffic Volume Across Nodes: In the Abilene dataset, mean traffic per node ranges from 4.46 to 640.29, while in GEANT it ranges from 86.75 to 15869.78, differing by over two orders of magnitude.

2) Presence of Outliers in the Dataset: Network anomalies such as interruptions and congestion can introduce outliers, which may distort statistical measures like means and variances [19].

To address these challenges, we apply RobustScaler [18] separately to each node. This node-wise normalization avoids overlooking low-traffic nodes, while RobustScaler's reliance on the median and IQR ensures robustness to outliers and

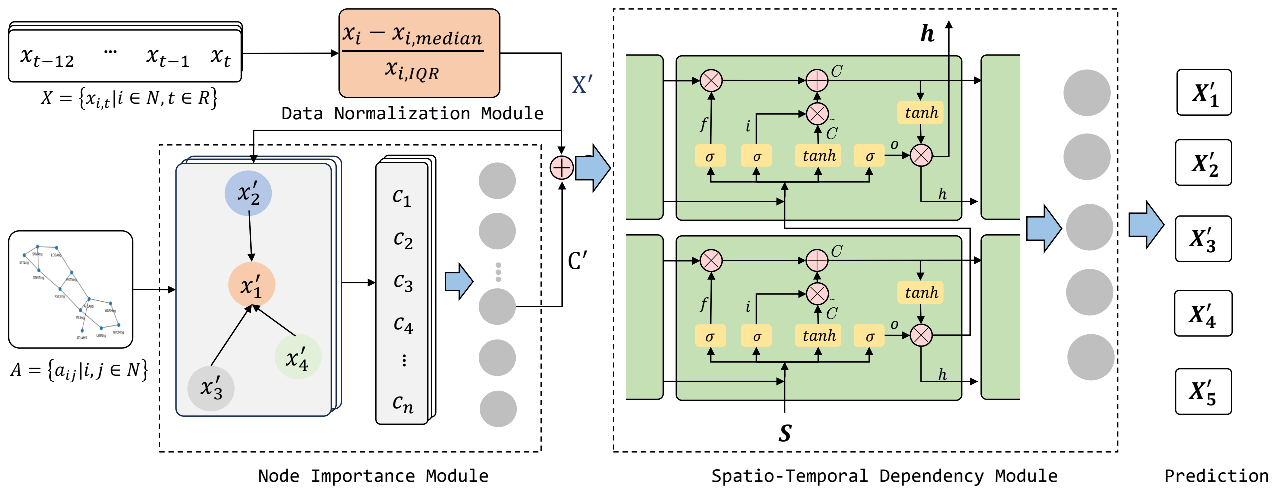

Fig. 2. The Overall Framework of the STGNI, Consists of Three Modules: the DNM (Data Normalization Module), the NIM (Node Importance Module), and the STDM (Spatio-Temporal Dependency Module).

distributional assumptions, thus outperforming StandardScaler and MinMaxScaler [20].

Taking node i in the dataset as an example:

$$x_i^{'} = \frac{x_i - x_{i,median}}{x_{i,IQR}} \tag{2}$$

Where $x_i = \{x_{i,t} | t \in R\}$ and $x_i^{'} = \{x_{i,t}^{'} | t \in R\}$ are the raw and normalized traffic series for node i, $x_{i,median}$ is the median, and $x_{i,IQR} = Q3 - Q1$ is the interquartile range.

Following the DNM, the historical traffic data $X = \{x_i | i \in N\}$ is transformed into normalized data $X^{'} = \left\{x_i^{'} | i \in N\right\}$. The normalized data is then used as the basis for model training in both NIM and STDM.

B. Node Importance Module.

The NIM computes node importance using a graph attention mechanism [21], which learns from the normalized data $X^{'} = \left\{x_i^{'} | i \in N\right\}$ and the adjacency matrix $A = \{a_{ij} | i,j \in N\}$ to capture relationships between nodes and their first-order neighbors. First, the attention coefficient e_{ij}, representing the importance of node j to node i, is computed as:

$$e_{ij} = \tanh\left(softplus\left(a^T\left[Wx_i^{'} || Wx_j^{'}\right]\right)\right) \tag{3}$$

Where W is a weight matrix, a is a parameterized vector, $\cdot T$ denotes matrix transposition, $||$ indicates concatenation, and both $tanh(\cdot)$ and $softplus(\cdot)$ are activation functions. The attention coefficients are masked by the adjacency matrix:

$$e_{ij} = \begin{cases} 0, a_{ij} = 0, \\ e_{ij}, a_{ij} = 1 \end{cases} \tag{4}$$

These coefficients are then normalized using the softmax function over all neighbors $j \in N_i$ of node i.

$$\alpha_{ij} = soft\max_{\mathrm{j}}(e_{\mathrm{ij}}) = \frac{\exp(e_{ij})}{\sum_{k \in N_i} \exp(e_{ik})} \tag{5}$$

Where α_{ij} denotes the attention weight between node i and node j, N_i represents the neighbors of node i in the graph. This mechanism focuses on information from its first-order neighboring nodes while dropping all structural information [21]. To improve generalization, a dropout layer is incorporated. The features of node i are then aggregated with those of its neighbors via weighted summation:

$$c_i = \tanh\left(softplus\left(\sum_{j \in N_i} \alpha_{ij} W x_j^{'}\right)\right) \tag{6}$$

The attention weights indicate the degree of influence neighboring nodes have on node i, with larger weights reflecting stronger relationships. Weighted aggregation thus enables effective integration of information from node i and its neighbors.

For parallelization, a multi-head attention mechanism concatenates outputs from K independent heads. The resulting coefficients c_i are transformed via a linear layer to obtain node importance $C^{'} = \left\{c_i^{'} | i \in N\right\}$, which is combined with the normalized data $X^{'} = \{x_i^{'} | i \in N\}$ as input to the subsequent module:

$$S = C^{'} + X^{'} \tag{7}$$

Where $S = \{s_{i,t} | i \in N, t \in R\}$ represents the final output.

C. Spatio-Temporal Dependency Module.

The STDM captures the spatio-temporal dependencies of node importance using a dual-layer LSTM, followed by a linear layer for network traffic prediction. The input sequence is denoted as $s_t = \{s_{t,i} | i \in N\}$. It is processed through the two LSTM layers to obtain the hidden state:

$$h_t = LSTM^{(2)}\left(LSTM^{(1)}(s_t)\right) \tag{8}$$

Where h_t represents the output of the dual-layer LSTM. The hidden state is then passed through a linear layer to generate predictions for future time step T:

$$X_T^{'} = sigmoid\left(W \cdot h_t + b_t\right) \tag{9}$$

Where $X_T^{'}$ denotes the predicted traffic, W and b_t are trainable parameters, and $sigmoid(\cdot)$ is activation function. In experiments, predictions are made up to five steps ahead $X_T^{'} = \{x_{i,t+T}^{'} \mid i \in N, t \in R, T \in \{1, 2, 3, 4, 5\}\}$.

V. Experiments.

A. Datasets.

SNDlib[1] serves as a repository of test instances tailored for Survivable Fixed Telecommunication Network Design [22]. From SNDlib, we select two frequently used dynamic traffic datasets: Abilene and GEANT. The XML files obtained from the website are first parsed to extract information about the source, target, and demand value. We then aggregate data across nodes, manually removing instances where all nodes have empty data, and fill missing values by averaging neighboring values. Following the DNM procedure outlined in Section IV-A, the final dataset is obtained and is available at https://github.com/JiaYanx/STGNI. The Abilene dataset contains traffic data from 12 nodes, recorded every 5 minutes, for a total of 16,382 valid data points. The GEANT dataset contains traffic data from 22 nodes, recorded every 15 minutes, for a total of 10,935 valid data points.

B. Settings.

We partition the dataset into training and testing sets in an 8:2 ratio according to temporal order, using the first 80% for training and the remaining 20% for testing to simulate future prediction. During model training, a fixed historical data window of 12 steps is used. To ensure fair comparison, each model is trained for 100 epochs with a batch size of 64. The hidden layer size is set to 32 for all models, except for LA-ResNet, where the LSTM hidden layer contains 64 nodes. The number of stacked layers (num_layers) is fixed at 2, and a dropout layer with a rate of 0.5 is applied. The optimizer uses a learning rate (lr) of 0.01. All models are trained three times following these settings.

C. Metrics.

For model evaluation, we use three metrics: MAE (Mean Absolute Error), MSE (Mean Squared Error), and RMSE (Root Mean Squared Error) [23, 24]. Smaller values indicate better prediction performance, while larger values reflect poorer accuracy.

[1] http://sndlib.zib.de/home.action

D. Baselines.

We select eight baseline methods divided into four groups: traditional statistical methods (ARIMA [4]); deep learning methods focusing solely on temporal features (RNN [25], GRU [26], TCN [27]); deep learning methods focusing solely on spatial features (GAT [12]); and deep learning methods capturing spatio-temporal relationships (LA-ResNet [28], GMAN [29], AGCRN [30]).

E. Results.

Table I and Table II respectively display the experimental results of each method on the Abilene and GEANT datasets. In these tables, "prediction" indicates the prediction of future T time steps of traffic data. Poor results are represented by "*" instead. All methods are evaluated using MAE, MSE, and RMSE, with smaller values indicating better performance.

1) **Analysis of STGNI's Performance on the Abilene Dataset.**

From Table I, STGNI consistently achieves the best results across most prediction steps, demonstrating its ability to effectively capture both spatial and temporal dependencies. Specifically, STGNI improves MAE, MSE, and RMSE by 23.08%, 5.13%, and 6.45% at step 1; 22.22%, 44.07%, and 29.87% at step 2; 17.86%, 18.64%, and 15.58% at step 3; and 15.63%, 5.33%, and 6.98% at step 5. Its only minor shortfall is at step 4, where GMAN slightly outperforms it.

Other models show weaker performance for different reasons. AGCRN and GMAN, although designed for spatio-temporal modeling, may not fit the specific characteristics of the Abilene dataset optimally, leading to sub-optimal predictions. LA-ResNet suffers from potential overfitting due to its larger number of parameters, while ARIMA cannot capture the nonlinear and complex temporal patterns in the data. Models focusing solely on temporal or spatial features fail to leverage the complementary information present in both dimensions, which limits their predictive accuracy.

2) **Analysis of STGNI's Performance on the GEANT Dataset.**

From Table II, it can be found STGNI achieves consistently superior results across all prediction steps, highlighting its ability to effectively capture both spatial and temporal dependencies. Specifically, STGNI improves MAE by 73.61%, 66.13%, 63.33%, 57.89%, and 53.70%; MSE by 88.00%, 78.33%, 75.44%, 67.92%, and 64.00%; and RMSE by 65.12%, 53.25%, 51.32%, 43.84%, and 40.00% for prediction steps 1 to 5, respectively.

Other methods perform worse for reasons similar to those discussed for the Abilene dataset. AGCRN, LA-ResNet, and ARIMA show sub-optimal performance for the same reasons. GMAN, although a spatio-temporal model, exhibits a decline in performance at larger prediction steps on this dataset, likely because it fails to extract new information as the prediction horizon increases. TCN, which focuses solely on temporal features, achieves the second-best performance.

TABLE I

RESULTS OF STGNI COMPARED WITH BASELINES ON THE ABILENE DATASET (BEST PERFORMANCE INDICATED IN BOLD).

Prediction	Method Metric	ARIMA	RNN	GRU	TCN	GAT	LA-ResNet	GMAN	AGCRN	STGNI
1	MAE	0.78	0.34	0.38	0.45	0.60	0.78	0.26	48.69	**0.20**
	MSE	3.59	0.78	1.01	0.65	2.97	3.43	0.39	*	**0.37**
	RMSE	1.44	0.85	0.95	0.78	1.66	1.78	0.62	*	**0.58**
2	MAE	0.78	0.38	0.31	0.52	0.60	0.86	0.27	29.64	**0.21**
	MSE	3.59	1.16	1.24	0.80	2.94	4.05	0.59	*	**0.33**
	RMSE	1.44	1.03	1.03	0.87	1.66	1.96	0.77	*	**0.54**
3	MAE	0.78	0.36	0.27	0.56	0.62	0.85	0.28	52.45	**0.23**
	MSE	3.59	1.29	1.08	0.99	3.00	4.06	0.59	*	**0.48**
	RMSE	1.44	1.08	0.97	0.97	1.67	1.96	0.77	*	**0.65**
4	MAE	0.78	0.31	0.30	0.57	0.61	0.85	**0.26**	51.14	**0.26**
	MSE	3.59	1.15	1.29	1.03	2.98	4.05	**0.56**	*	0.68
	RMSE	1.44	1.02	1.08	0.99	1.67	1.95	**0.75**	*	0.78
5	MAE	0.78	0.32	0.31	0.67	0.61	0.85	0.32	51.53	**0.27**
	MSE	3.59	1.13	1.13	1.86	2.99	4.06	0.75	*	**0.71**
	RMSE	1.44	1.01	1.01	1.32	1.67	1.96	0.86	*	**0.80**

TABLE II

RESULTS OF STGNI COMPARED WITH BASELINES ON THE GEANT DATASET (BEST PERFORMANCE INDICATED IN BOLD).

Prediction	Method Metric	ARIMA	RNN	GRU	TCN	GAT	LA-ResNet	GMAN	AGCRN	STGNI
1	MAE	3.81	3.68	3.13	0.72	0.79	1.46	0.77	928.53	**0.19**
	MSE	*	24.08	41.68	0.75	0.85	2.48	0.90	*	**0.09**
	RMSE	*	4.79	5.84	0.86	0.92	1.57	0.95	*	**0.30**
2	MAE	3.82	2.71	4.20	0.62	0.77	2.34	6.87	809.03	**0.21**
	MSE	*	13.28	54.43	0.60	0.82	5.92	47.69	*	**0.13**
	RMSE	*	3.59	6.75	0.77	0.91	2.42	6.91	*	**0.36**
3	MAE	3.82	5.08	5.54	0.60	0.71	2.53	7.31	611.52	**0.22**
	MSE	*	57.33	106.09	0.57	0.73	6.87	280.73	*	**0.14**
	RMSE	*	7.45	9.38	0.76	0.85	2.62	16.76	*	**0.37**
4	MAE	3.82	12.79	5.19	0.57	0.75	2.52	5.46	885.42	**0.24**
	MSE	*	996.43	103.79	0.53	0.80	6.78	203.94	*	**0.17**
	RMSE	*	24.26	9.29	0.73	0.89	2.60	14.28	*	**0.41**
5	MAE	3.82	2.88	5.47	0.54	0.71	2.37	2.37	761.72	**0.25**
	MSE	*	17.15	132.86	0.50	0.72	6.00	355.73	*	**0.18**
	RMSE	*	3.93	9.84	0.70	0.85	2.45	18.86	*	**0.42**

F. Ablation Studies

We conduct an ablation study to evaluate the contributions of NIM and STDM within STGNI. Here, STGNI(ALL) includes DNM, NIM, and STDM; STGNI(NIM) includes DNM and NIM; STGNI(STDM) includes DNM and STDM.

1) Effectiveness of NIM: STGNI (STDM) shows dataset-dependent performance. On Abilene, it achieves results comparable to STGNI (ALL), with MAE, MSE, and RMSE improvements of 23.08%, 30.19%, and 17.14% at step 1, gradually decreasing over steps 2–5. On GEANT, STGNI (STDM) underperforms compared to STGNI (NIM), indicating node importance is more critical. Nevertheless, combined with NIM in STGNI (ALL), STDM contributes to substantial improvements across steps 1–5, with MAE improvements ranging from 93.24% at step 1 to 84.76% at step 5.

2) Effectiveness of STDM: Comparing STGNI (NIM) with STGNI (ALL), we observe significant performance drops across both datasets, highlighting the importance of STDM in capturing node-specific contributions. On the Abilene dataset, including STDM improves MAE, MSE, and RMSE across prediction steps 1–5, with the largest gains at step 1 (66.67%, 87.54%, 65.06%) and gradual decreases for later steps. Similarly, on the GEANT dataset, MAE, MSE, and RMSE improve consistently at steps 1–5, with step 1 improvements of 75.95%, 89.41%, and 67.39%, indicating that modeling node importance substantially enhances prediction accuracy.

VI. CONCLUSION

In this work, we proposed STGNI for network traffic prediction, integrating DNM, NIM, and STDM to jointly address data variability, node importance, and spatio-temporal dependencies. Experiments on the Abilene and GEANT datasets against eight baselines demonstrated that STGNI consistently achieves superior performance, validating the effectiveness of

TABLE III

RESULTS OF ABLATION STUDIES ON ABILENE AND GEANT DATASETS (BEST PERFORMANCE INDICATED IN BOLD).

Prediction	Dataset	Abilene			GEANT		
	Method Metric	STGNI (NIM)	STGNI (STDM)	STGNI (ALL)	STGNI (NIM)	STGNI (STDM)	STGNI (ALL)
1	MAE	0.60	0.26	**0.20**	0.79	2.81	**0.19**
	MSE	2.97	0.53	**0.37**	0.85	28.05	**0.09**
	RMSE	1.66	0.70	**0.58**	0.92	4.76	**0.30**
2	MAE	0.60	0.24	**0.21**	0.77	4.56	**0.21**
	MSE	2.94	0.64	**0.33**	0.82	51.46	**0.13**
	RMSE	1.66	0.74	**0.54**	0.91	5.95	**0.36**
3	MAE	0.62	0.24	**0.23**	0.71	2.37	**0.22**
	MSE	3.00	0.50	**0.48**	0.73	19.4	**0.14**
	RMSE	1.67	0.67	**0.65**	0.85	4.06	**0.37**
4	MAE	0.61	**0.26**	**0.26**	0.75	2.88	**0.24**
	MSE	2.98	**0.64**	0.68	0.80	38.91	**0.17**
	RMSE	1.67	**0.77**	0.78	0.89	6.12	**0.41**
5	MAE	0.61	0.28	**0.27**	0.71	1.64	**0.25**
	MSE	2.99	0.82	**0.71**	0.72	8.82	**0.18**
	RMSE	1.67	0.85	**0.80**	0.85	2.82	**0.42**

both the overall framework and its individual modules. Nevertheless, we observed a slight performance drop at the 4-step prediction horizon on the Abilene dataset. This highlights the need for further exploration of long-term prediction strategies, which we plan to pursue in future work.

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